MILITARY SPECIFICATION

RESISTOR, FIXED, FILM, HIGH RELIABILITY
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-87254, dated 1 October 1990, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

Following 2.1.1, add:

"2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NASA Reference Publication 1124 - Outgassing Data for Selecting Spacecraft Materials.

(Application for copies should be addressed to the NASA/Parts Project Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ASTM E 595 - Standard Test Method for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment.

JSC 08962 - Compilation of VCM Data of Nonmetallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services. "

2.2, delete and substitute the following:

"2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-23T12:22Z
Check the source to verify that this is the current version before use.

3.4.3, Second sentence, delete "C31."

Add the following new paragaraph:

"3.4.3.1 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat (see 6.5). Use of tin-lead (Sn-Pb) finishes are acceptable provided that minimum lead content is 3 percent."

Following 3.4.3, add:

"3.4.3.1 Solder dip (retinning) leads. The manufacturer (or his authorized category B or C distributor) may solder dip/retin the leads of product supplied to this specification provided the solder dip process has been approved by the qualifying activity. The manufacturer (or his authorized category B or C distributor) shall maintain a solder purity in accordance with table IIIa during the tinning process.

TABLE IIIa. Contamination limits.

Contamination	Tinning percent by weight 1/
Copper	.75
Gold	.50
Cadmium	.01
Zinc	.008
Aluminum	.008
Antimony	.50
Iron	.02
Arsenic	.03
Bismuth	.25
Silver	.75
Nickel	.025

1/ This is a fixed percentage by weight of the solder.

"3.4.3.2 Qualifying activity approval. Approval of the solder dip process will be based on one of the following options:

a. When the original lead finish qualified was hot solder dip lead finish 52 of MIL-STD-1276. (NOTE: The 200 microinch thickness is not applicable.) The manufacturer shall use the same solder dip process for retinning as is used in the original manufacture of the product.

b. When the lead originally qualified was not hot solder dip lead finish 52 of MIL-STD-1276 as prescribed in 3.4.3.2a, approval for the process to be used for solder dip shall be based on the following test procedure:

(1) Thirty samples of any resistance value for each style and lead finish are subjected to the manufacturing's solder dip process. Following the solder dip process, the resistors are subject to the direct current (dc) resistance test and other group A electricals. No defects are allowed. (NOTE: If hermetic seal testing is required in group A, these tests would also be performed.) No defects are allowed.

(2) Ten of the 30 samples are then subjected to the solderability test. No defects are allowed.

(3) The remaining 20 samples are subjected to the resistance to solder heat test followed by the moisture resistance test (or hermetic seal test if the device is hermetically sealed). No defects are allowed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T12:22Z
Check the source to verify that this is the current version before use.

"3.4.3.3 Solder dip/retinning options. The manufacturer (or authorized category B or C distributor) may solder dip/retin as follows:

a. After the 100 percent group A screening tests: Following the solder dip/retinning process, the electrical measurements required in group A, subgroup 1, 100 percent screening tests shall be repeated on 100 percent of the lot (NOTE: The manufacturer may solder dip/retin prior to the 100 percent electrical measurements of the group A, subgroup 1 tests). The percent defective allowable (PDA) (see 4.6.1.2.1) for the electrical measurements shall be as for the subgroup 1 tests. (NOTE: If hermetic seal is required in group A, subgroup 1, this test must be performed after the solder dip/retinning process or it will have to be repeated.)

b. As a corrective action, if the lot fails the group A solderability tests, the lot may be retinned no more than two times. The lot after retinning shall be 100 percent screened for group A electrical requirements (dc resistance) any parts failing (lot not exceeding PDA for group A, subgroup 1, see 4.6.1.2.1) these screens shall not be supplied to this specification. If electrical failures are detected after the second retinning operation exceeding 1 percent of the lot, the lot shall not be supplied to this specification.

c. After the group A inspection has been completed: Following the solder dip/retinning process, the electrical measurements required in group A, subgroup 1, 100 percent screening tests shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be as for the subgroup 1 tests (see 4.5c). (NOTE: If hermetic seal is required in the group A, subgroup 1 tests, this test shall be repeated.) Following these tests, the manufacturer shall submit the lot to the group A solderability tests as specified in 4.7.6. ."

PAGE 7

3.6, delete "E = PR", and substitute:

"E = √PR"

PAGE 8

Following 3.22, add:

"3.22.1 Thermal vacuum outgassing (qualification only). Materials used shall not release greater than 1.0 percent total mass loss (TML) or 0.1 percent collected volatile condensable material (CVCM) when tested as specified in 4.7.16. Data listed in NASA Reference Publication 1124, revised, or JSC 08962 may be used in lieu of actual test data for applicable materials."

PAGE 12

TABLE IV, subgroup 5, delete all applicable information and substitute as follows:

"	Subgroup 5				
	Solderability	3.12	4.7.6	5 samples	
	Subgroup 6				
	Destructive physical analysis	3.21	4.7.15		
	Hermetically sealed			5 samples	
	Nonhermetically sealed			8 samples	
				0 failures	"

PAGE 13

4.6.1.2.1.1, at the end of the paragraph, add:

"Units that are removed from the lot as being outside of ±3 standard deviations and pass the change in resistance limit shall not be counted when calculating the 3 percent total rejects allowed from subgroup 1."

Source: https://assist.dla.mil -- Downloaded: 202610-23T12:22Z
Check the source to verify that this is the currentversion before use.

PAGE 14

TABLE V, subgroup 1, delete:

"Solderability" and all applicable information.

PAGE 18

4.7.4a, delete and substitute:

"a. Measuring apparatus: Different types of measuring test equipment (multimeters, bridges, or equivalent) are permitted to be used on the initial and final readings of this test, provided the equipment is the same style, model, or if it can be shown that the performance of the equipment is equivalent or better."

PAGE 21

4.7.12d, delete in its entirety.

PAGE 24

Following 4.7.16, add:

"4.7.16.1 Thermal vacuum outgassing (qualification only) (see 3.22.1). Materials used shall be tested in accordance with ASTM E 595. Samples to be tested shall be identical to materials used in the qualification lot."

PAGE 25

Add the following new paragraph:

"6.5 Tin plated finishes. Tin plating is prohibited (see 3.4.3.1) since it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter, refer to ASTM B545 (Standard Specification for Electrodeposited Coating of Tin)."

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 19
NASA - NA

Review activities:
Army - AR, AT, AV, ME
Navy - AS, CG, MC, OS
Air Force - 17, 85, 99
DLA - ES

Preparing activity:
Air Force - 19

Agent:
DLA - ES

(Project 5905-1323)

Source: https://assist.dla.mil -- Downloaded: 202610-23T12:22Z
Check the source to verify that this is the current version before use.